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MILITARY SPECIFICATION

MACHINING CENTERS, VERTICAL, SINGLE SPINDLE, COMPUTER NUMERICAL CONTROL (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers computer numerically controlled (CNC), single vertical spindle type machining centers, with an automatic tool changer and other components or with equipment and accessories specified in Section 3. The requirements set forth in this specification and the completed ordering data (paragraph 6.2) establishes the Government's minimum requirements for each procurement.

1.2 Classification. The machining centers covered by this specification have three (CNC) controlled axes and shall be of the following sizes and styles. The sizes and styles to be furnished shall be as specified (see 3.5 and 6.2).

Beneficial commments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3408
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	Spindle motor horsepower	Mill capacity Cubic inches/minute (SAE 1020 Steel)	CNC controlled travel X-Y-Z, inches
Size 1	5	5	30 x 15 x 8
Size 2	7	7	30 x 15 x 10
Size 3	7	7	50 x 26 x 10
Size 4	10	10	40 x 20 x 12
Size 5	10	10	50 x 26 x 12
Size 6	10	10	80 x 26 x 12
Size 7	15	15	54 x 28 x 14
Size 8	20	20	54 x 28 x 24
Size 9	40	40	80 x 40 x 40

Style A - Tool magazine capacity of 15 or more tools
Style B - Tool magazine capacity of 20 or more tools
Style C - Tool magazine capacity of 30 or more tools
Style D - Tool magazine capacity of 40 or more tools
Style E - Tool magazine capacity of 60 or more tools
Style F - Tool magazine capacity of 80 or more tools
Style G - Tool magazine capacity of 100 or more tools

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

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MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk (ATTN: Mr. Stephen Lowell), Building 4D, 700 Robbins Avenue Philadelphia, PA 19111-5094.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears. (DOD adopted)

AGMA 2000-A88 - Gear Classification and Inspection Handbook. Tolerances and Measuring methods for Unassembled Spur and Helical Gears (including Metric Equivalents). (DOD adopted)

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AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth. (DOD adopted)

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, and Tongues. (DOD adopted)

ANSI B5.50 - "V" Flange Tool Shanks for Machining Centers with Automatic Tool Changers. (DOD adopted)

ANSI B11.8 - Drilling, Milling, and Boring Machines, Safety Requirements for the Construction, Care, and Use. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

ANSI B93.114 - Pneumatic Fluid Power-Systems-Standard for Industrial Equipment and Machine Tool Applications.

ANSI/ISO 8860/1 - Dimensional, Physical and Magnetic Characteristics.

ANSI/ISO 8860/2 - Track Format.

ANSI/ISO 9293 - Volume and File Structure of Flexible Disk Cartridges for Information Interchange.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227 - One-Inch Perforated Tape. (DOD adopted)

RS-232 - Interface between Data Terminal Equipment and Data Circuit Terminating Equipment Employing Serial Binary Data Interchange. (DOD adopted)

RS-244 - Character Code for Numerical Machine Control Perforated Tape. (DOD adopted)

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RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machines. (DOD adopted)

RS-274 - Interchangeable Variable Block Data Contouring, Format for Positioning, and Contouring/Positioning Numerically Controlled Machines. (DOD adopted)

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape. (DOD adopted)

RS-491 - Interface Between a Numerical Control Unit and Peripheral Equipment Employing Asynchronous Binary Data Interchange Over Circuits Having EIA-423-A Electrical Characteristics.

RS-494 - 32 Bit Binary CL Exchange (BCL) Input Format for Numerically Controlled Machines.

(Application for copies should be addressed to the Electric Industries Association, 2001 Eye Street, N.W., Washington. DC 20006-1899.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems. (DOD adopted)

NEMA ICS 3 - Industrial Systems. (DOD adopted)

NEMA MG 1 - Motors and Generators. (DOD adopted)

NEMA MG 7 - Motion/Position Control Motors and Controls. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2), one complete machining center shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The machining center shall be new (not a prototype) and one of the manufacturer's current models. Its design shall include features and components necessary for maintaining alignment and accomplishing drilling, tapping, milling, and boring performance required herein. The machine shall have a vertical axis spindle assembly housed within a machining head which shall be supported by a column and floor mounting base. The base shall also support a saddle and/or worktable which shall be arranged to support and secure the workpiece for machining operations. The worktable and/or column shall have slide motion (travel) as necessary for altering the relationship of the spindle with the workpiece in the X and Y axes. Unless otherwise specified (see 6.2), Z axis movement shall be accomplished by either a machining head or machining head and quill, sliding vertically along the column ways to and from the worktable or with the machining head affixed to the column and by a quill sliding within the head to and from the worktable. In the latter case, the worktable shall also be vertically adjustable. Tools, toolholders and adapters shall be inserted and removed from the spindle by a tool changer with a storage magazine. Tool change, spindle operation, and motion in the X, Y, and Z axes shall be directed by a computer numerical control (CNC) system. The machine and its numerical control system shall constitute a completely functional system with built-in operations programmed by coded punch tape and/or other devices. The machine functions shall be controlled manually by operator control devices, semiautomatically by manual data input devices, and automatically from part program data input. All parts of the machine or system that are subject to wear, breakage, or distortion shall be accessible for adjustment, replacement and repair. The design and construction of the machine shall be in accordance with either the U.S. Customary System of Units (US) or the International System of Units (SI).

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to

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become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machining center and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, electrical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.8. If a conflict arises between 29 CFR 1910 and ANSI B11.8, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Environmental protection. The machining center shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. During the operation of the machine at its final location, it shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract regulating the emission of materials hazardous to the environment. Specific environmental protection requirements exceeding the established EPA and OSHA regulations as required herein shall be as specified by the procuring activity (see 6.2).

3.2.7 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter and a low lubrication level warning light that is in clear view of the operator's normal work station. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.8 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the

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interchangeability of respective parts between end items of the same model. All replacement parts shall be interchangeable with their respective original parts and fit/function with mating components.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. Any material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations.

3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2000-A88 and AGMA 2001-B88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461.

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The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machining center shall consist of not less than a base, column, saddle, worktable, machining head, spindle assembly, tool changer, coolant system, electrical system, and computerized numerical control system.

3.4.1 Base. The base shall be a casting of iron, iron alloy, or composite polymer concrete or a fabricated steel weldment, ribbed and braced to minimize distortion and deflection. The base shall form a foundation for the entire machine and be integrally constructed with the column or be separately constructed with the column machined, securely attached, and in either case, be sufficiently rigid to maintain mutual component alignment for assuring accuracy required herein. The base shall have guideways that are proportioned to fully support the saddle and/or column and worktable throughout its full range of travel. The base shall have means for leveling the machine on a flat mounting surface. Hydraulic or coolant fluid reservoirs may be housed within the base or be supplied as separate external units. Reservoirs shall have means for indicating fluid levels. The base shall have access or removable plates as necessary for servicing components housed within.

3.4.2 Column. The column shall be a casting of iron, iron alloy, or composite polymer concrete or a fabricated steel weldment of box-like construction, internally ribbed and braced to minimize distortion and deflection under normal tool loads. The column shall be designed to support and accommodate the machining head and have guideways to facilitate vertical movement of the head or worktable. If column movement is used to provide either X or Y axis movement, the column shall be precision fitted to the guideways of the base. Column slides and screws shall be shielded and protected from foreign contamination by either telescoping or curtain type way covers and way wipers.

3.4.3 Saddle. The saddle slide shall be a casting of iron, iron alloy or a fabricated steel weldment, rigidly constructed and precision fitted to the guideways of the base. The saddle shall also have guideways for supporting and traversing the worktable throughout its full range of travel. Guideways of the saddle and all other components of the machine shall be designed with friction characteristics that will permit precise control of slide movement without sticking or overshooting. The guideways of the machines shall be hardened and ground having a hardness not less than Rockwell C-60. Mating ways shall have means for holding them in alignment to prevent cutting forces from separating slide surfaces or distorting feed screws. Means such as servo-locks shall be provided that automatically secure slide surfaces in place immediately upon completing numerical control-directed moves.

3.4.4 Worktable. The worktable shall be a casting of iron, or iron alloy, internally ribbed and proportioned to support and withstand maximum capacity workpiece weights and tool thrust loads. The worktable shall be precision

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fitted to the ways of the saddle and be the compound type, arranged to provide workpiece travel in X and Y axes. The top of the worktable shall be machine finished for a work mounting surface. The worktable shall have a coolant trough or similar means for draining spent coolant into its reservoir. Unless otherwise specified (see 6.2), the number of T-slots shall be the manufacturer's standard. The size and configuration of T-slots shall be in accordance with ANSI B5.1. When required, the worktable shall be furnished with a subplate of the characteristics specified (see 6.2).

3.4.5 Machining head. The machining head shall be a casting of iron or iron alloy or a fabricated steel weldment designed and proportioned to house and support the quill (as required), spindle assembly, and spindle rotational drive with motor and related components necessary for accomplishing drilling, tapping, milling, and boring. The machining head shall hold the spindle in axial alignment perpendicular to the work mounting surface of the worktable and shall be designed to prevent distortion of the head due to torsional forces of the drives. The Z axis movement shall be servo-driven and fully controlled by the numerical control system. The machine shall have provisions at the upper and lower limits to prevent excessive overtravel in the Z axis.

3.4.5.1 Spindle assembly. The spindle shall be of alloy steel hardened and ground to a finish not rougher than 32 microinches. The spindle shall be supported and held in axial and radial alignment by ball or roller bearings. The spindle shall be mounted in the machining head with its bearings designed and arranged to compensate for thermal expansion. The spindle shall have positive means for accepting, retaining, and releasing tools and tool holders inserted by either the automatic tool changer or the operator. Unless otherwise specified (see 6.2), nose, bores, and other characteristics of the spindle shall be in accordance with ANSI B5.50. The spindle shall automatically stop rotation for tool changes, and shall be equipped to automatically orient to a preprogrammed position enabling the automatic tool changer to insert preprogrammed tools from the tool magazine. The spindle shall be capable of both the clockwise and counterclockwise directions of rotation for accomplishing the full range of spindle speeds specified herein. The spindle shall be driven by a variable speed electric motor. The machine shall have overload protection which automatically stops the movement of all slides and the rotation of the spindle should it stall from overload.

3.4.6 Machine ways. The machine ways shall be protected wherever possible to exclude chips, dirt, coolant, and other hazards that may cause damage or result in inaccurate functioning of any members. Protection of the machine ways shall be either telescoping or curtain type way covers and way wipers. The machine ways shall be hardened and ground with a minimum hardness of 60 Rockwell "C". Linear slide movement over the ways shall be supported by either precision hardened and ground bearing roller packs or non-metallic liners having a low coefficient of friction, with excellent slip-stick and dampening properties.

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3.4.7 Tool changer. The machine shall have an automatic tool changer with a magazine that is capable of storing not less than the tool capacity required for the style machine specified in the contract or order. The magazine shall be of the rotary drum design or of any other closed loop recirculating configuration with the capability of presenting a preprogrammed tool to the tool changer for exchange with a tool held in the spindle. Tool change shall be accomplished by means of a power-actuated mechanism which shall either interchange the tool in the spindle with a tool selected from the tool magazine or remove the tool from the spindle then insert the tool from the magazine. The magazine shall permit convenient loading and unloading of tools, tool holders, and adapters during operational set up. The tool changer shall be arranged for random tool selection which shall include the capability of reusing a tool that has been returned to the magazine after a program operation.

3.4.8 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.9 Pneumatic system. The pneumatic system shall conform to the requirements of ANSI B93.114M.

3.4.10 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.11 Electrical requirements. The machine shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), the machine shall be designed to operate on dual 230/460-volt, 3-phase, 60 Hertz (Hz) input power. The machine shall be initially wired for operation on the voltage (230 or 460 volts) specified by the procuring activity (see 6.2).

3.4.11.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. All motors (except axis motors, see 3.4.11.2) shall conform to the requirements of NEMA MG-1. Unless otherwise specified (see 6.2), motors shall be alternating-current (AC), 60-Hertz (Hz), or direct-current (DC) industrial -type motors with dripproof enclosures. Each motor control shall conform to the requirements of NEMA ICS-1 and NEMA ICS-3.

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3.4.11.2 Axis motor(s) and motor controls. Unless otherwise specified (see 6.2), axis motors shall be DC or AC, permanent magnet, servo-type motors with ball or roller bearings of the sealed and permanently lubricated type. The motor enclosures shall be totally enclosed nonventilated, (TENV) with provision for mounting a cooling fan and motor. Unless otherwise specified (see 6.2), the axis motor controls shall be silicon controlled rectifier (SCR) or pulse width modulated (PWM) solid-state type controls. Each motor and control shall conform to the requirements of NEMA MG-7.

3.4.12 Worklight. Unless otherwise specified (see 6.2) a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.13 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4.14 Computer numerical control (CNC). The machining center shall have a fully automatic solid-state soft-wired, integrated circuit, microprocessor based-type computer numerical control. The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, axes movement, spindle operation, tool change, and other part program directed functions for machining centers of the type specified herein. Controlled axes shall be identified in accordance with EIA Standard RS-267. Control features shall include linear and circular interpolation, manual data input (MDI) simultaneous control of three linear axes, programmable interface, buffer storage, fixed cycles, part program storage, part program and buffer edit capability, and control diagnostics to at least board level. The MCU shall direct machine functions from command data stored in memory. The data stored in memory shall be input by the media specified herein or manual data input (MDI) from a full ASCII keyboard. The control shall initiate a halt or error signal should a fault condition occur in the control. All necessary executive program routines shall be furnished in the form of 1-inch eight-channel binary coded decimal punched tape in conformance with EIA Standard RS-227 or 3-1/2 inch diskettes for controls having volatile memories. When executive programs are supplied in the form of erasable programmable read only memory EPROM(s), complete documentation sufficient to interpret and duplicate shall be provided or one additional EPROM chip shall be provided by the manufacturer. A line voltage of $\pm$10 percent from normal shall not adversely affect the numerical control system. The control shall be capable of functioning in ambient temperatures ranging from 50^{0} to 120^{0} Fahrenheit (F) and humidity ranging from 5 to 95 percent noncondensing. The control shall automatically shut down when internal operating temperatures exceed safe operating limits. Control

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resolution shall be 0.0001 inch or 0.001mm for each linear axis and not more than 0.001 of a degree for each rotary axis. The control shall also provide the following:

a. A 9-, 12-, or 14-inch monochrome or multi-color cathode ray tube (CRT) shall be provided.

b. Unless otherwise specified (see 6.2), a photoelectric tape reader shall be provided that is capable of reading a minimum of 300 characters per second with standard feed and take up spools 5-1/4 inches in diameter.

c. Automatic tape code recognition, in conformance with EIA Standards RS-244 and RS-358.

d. Decimal point programming.

e. Preparatory functions (G-codes) and miscellaneous functions (M-codes) in accordance with EIA Standard RS-274

f. Programmable software travel limits, when specified (see 6.2).

g. Zone inhibit programming, when specified (see 6.2).

h. Axis scaling, when specified (see 6.2).

i. Polar coordinate programming, when specified (6.2).

j. Direct revolutions per minute (RPM) programming.

k. Radius programming, when specified (see 6.2).

l. Programmable dwell.

m. Absolute/incremental input.

n. Inverse time feed rate, when specified (see 6.2).

o. Parametric programming, when specified (see 6.2).

p. Inch/metric switchable data input.

q. Axis inversion.

r. Helical interpolation, when specified (see 6.2).

s. Plane selection.

t. Coordinate system plane rotation, when specified (see 6.2).

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u. Feedrate programming inches per minute/millimeters per minute (IPM/MMPM).

v. Fixed cycles, G80 series.

w. Dry run, no motion test with graphics.

x. Spindle speed and axis feedrate override.

y. EIA Standards RS-232 and RS-449 serial program data input interface, unless otherwise specified (see 6.2).

z. Software for additional rotary axis, positioning or contouring type, when specified (see 6.2).

aa. Cutter diameter compensation and tool length offsets.

bb. Automatic acceleration/deceleration all axes.

cc. Three- or four- axis simultaneous control, when specified (see 6.2).

dd. Tool life management, when specified (see 6.2).

ee. Linear axis jog for each axis, continuous and incremental modes.

ff. Leadscrew pitch error compensation.

gg. Reversal error compensation.

hh. Axis error compensation.

ii. Fixture error compensation.

jj. Multiquadrant circular interpolation, when specified (see 6.2).

kk. Standby battery power for controls with volatile memories shall be a minimum of 72 hours, unless otherwise specified (see 6.2).

ll. The control shall be capable of storing in memory not less than 100 feet of part programs. Part program storage shall be one of the following as specified (see 6.2): solid-state, magnetic bubble memory, Winchester hard disk/flexible disk drive combination units or single/or double flexible disk drive units.

mm. Buffer storage: The buffer shall store transferred command data from the specified input media to internal storage without delaying the next incoming command and without interrupting the machine functions.

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nn. Graphics software shall be provided which shall allow for on-screen display of part geometry and tool path verification. Software functions shall be menu driven with interactive conversational prompting to allow for programming at the MCU through the MDI keyboard.

oo. Cycle start/feed hold.

pp. Program edit, change, add and block, delete.

qq. Sequence number search forward and reverse.

rr. Reference zero, (all axes home position).

ss. Floating zero.

tt. Key lock for program edit.

uu. Plus and minus programming.

vv. Programmable macros.

ww. Thread cutting capabilities.

3.4.14.1 Postprocessor. When specified (see 6.2), the system shall be furnished with postprocessor programs to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall be provided for converting specified programming language (see 6.2) to the form and format required to operate the machine tool and control. The postprocessor shall be written for use on the type computer specified (see 6.2). Documentation sufficient to implement and utilize the postprocessor shall be provided for computer compatibility verification and validation at least 30 days prior to delivery of the machining center.

3.4.14.2 Tape punch unit. When specified (see 6.2), the system shall have a tape punch unit for producing punched tapes from edited programs of tape being input and for programs stored in data memory. The unit shall have a plug receptacle and circuitry that is compatible with the tape punch output of the system, punching both paper and mylar tapes of dimensions, characteristics and code in accordance with EIA Standards RS-244, RS-274 and RS-358.

3.4.14.3 Diskette input. When specified (see 6.2), a portable diskette reader/writer shall be provided with all the hardware and software necessary and interface between the control and the reader/writer. The reader/writer shall receive and store part program data from the control unit and be capable of loading stored programs to the memory of the control unit. Part program storage shall be on a 3-1/2 inch diskette. The diskette shall meet the

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requirements of ANSI/ISO 8860/1, ANSI/ISO 8860/2, and ANSI/ISO 9293. The communication between the reader/writer shall be by means of EIA Standard RS -232 communication ports.

3.4.14.4 Binary Cutter Location (BCL). When 32 Bit Binary Cutter Location (BCL) Exchange Input for Numerically Controlled Machines is specified (see 6.2), the input media shall be a 3-1/2 inch diskette and shall meet the requirements of ANSI/ISO 8860/1, ANSI/ISO 8860/2, and ANSI/ISO 9293. The control shall recognize and act upon command data formatted in accordance with EIA Standard RS-494. Section 10.1 shall apply to all BCL controls. Specific function set(s) required shall be provided as specified (see 6.2). Manufacturers providing functions or features that can not be addressed by command codes currently covered by EIA Standard RS-494 shall apply to the Electronic Industries Association, IE-31 Committee on Numerical Control Systems and Equipment for an extension of the standard codes to cover the new features or functions. A diskette reader shall be provided. The reader shall have the capability to perform a cyclic redundancy check of the input data and be capable of reading not less than 15,000 characters per second.

3.4.14.5 Punched tape data input. Unless otherwise specified (see 6.2), the input media for the control shall be 1-inch, eight-channel punched tape conforming to EIA Standard RS-227. The data input format and codes shall comply with EIA Standards RS-274 and both RS-244 and RS-358.

3.4.14.6 Rotary table, "A, B or C" axis. When specified (see 6.2), the control system shall be configured for either an "A, B or C" rotary axis and the machine shall be furnished with a rotary table. The rotary table shall be of the contouring or positioning type as specified (see 6.2). The rotary table shall be either an integral component of the machine or an attachment to the worktable. The top of the rotary table shall have T-slots for workpiece attachment. The size and configuration of the T-slots shall be in accordance with ANSI B5.1. The rotation of the specified rotary axis shall be parallel to its respective axis and conform to EIA Standard RS-267. The control shall direct rotation of the rotary table to any of the program selectable (degree) positions specified (see 6.2).

3.4.14.7 Operator control panel. Unless otherwise specified (see 6.2), the control unit shall have a control panel which provides at least the following control functions: emergency stop, cycle start, optional stop, mode selection, feed hold, feedrate override, spindle speed override, jog, jog direction, and EOB Stop.

3.4.14.7.1 Pendant control. When specified (see 6.2) a pendant control shall be furnished as an operator control panel. The pendant control shall be connected to the machine to enable the operator to control machine functions while in a close observation position. Unless otherwise specified (see 6.2), the pendant control shall be capable of the same functions as stated for the operator control panel in paragraph 3.4.14.7.

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3.4.14.8 Programmable interface. A software program for accommodating the softwired interface between the numerical control and machine shall be provided either in the control as nonvolatile memory or as a separate program that is compatible with the input media specified herein by the procuring activity.

3.4.14.9 Maintenance diagnostics system. Maintenance diagnostics software shall be furnished either in the control as nonvolatile memory or as a separate program that is compatible with the input media specified herein by the procuring activity. The diagnostic software shall test, exercise, and display failures, at least to board level.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I for the machine specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. Physical size of the machine shall be limited, when specified (see 6.2). The physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table III.

3.7 Performance. The machine and computer numerical control system combination shall be capable of performing as specified herein. In addition to the test specified in 4.8.4, the machine shall be capable of performing drilling, tapping, milling, and boring to accomplish the capacities of table I while meeting the accuracies of table II.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.10 Spare parts. A required spare parts list, as recommended by the manufacturer, shall be furnished to the procuring activity 90 days prior to machine installation. The procuring activity shall decide on the type and number of spare parts to be procured.

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3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below:

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below:

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Type of fluid (SAE or military or Federal specification number)
Type of filter

3.12 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein.

3.13 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 that all technical data ordered shall be written in the English language.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of

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all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see

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6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° and 20 to 95 percent relative humidity.

4.8.2 Operational test. Each machine shall be operated under numerical control for a period of 8 hours without load. The numerical control program shall cycle each axis through its entire range, addressing all speeds, feeds, direction of rotation, and standard control functions specified in this document. Proper operation of all machine functions, controls, motors, adjusting mechanisms, tool changes and accessories shall be verified during the test period. The numerical control system shall be further tested to verify proper operation of MDI, program edit, graphics and system diagnostics. Should a malfunction occur, it shall be corrected and the operational test repeated until a full 8 hours of running time is completed without failure.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance tests given in 4.8.4 and 4.8.4.1. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.4 Performance tests. While using one or more test setups devised by the supplier, the machine shall be tested to verify its ability to perform machining operations under numerical control for meeting the accuracy tolerances of table II. The test piece(s) may be, at the supplier's option. Steel, cast iron or aluminum and the tool(s), spindle speed and feed rate for each test shall be selected by the supplier as the most suitable for the particular operation being performed. The test piece(s) shall present a work surface normal to the tool point and approximately centered with reference to the work area in the XY plane. In the milling and contouring test, successive cuts may be made inside the outlines of previous cuts and roughing cuts may be made in the various cuts with excess metal removed as desired to make measurement convenient.

4.8.4.1 Axis positioning and repeatability tests. Starting at positions other than the extremes of axis travel, two identical movements of each controllable linear axis shall be programmed and executed in each direction. Programmed span shall be not less than 7 inches, and each digit of starting and end points shall be other than zero with respect to reference zero. Axis feedrate shall be 80 percent of maximum feedrate. Absolute positioning and

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positioning repeatability errors shall not exceed the permissible tolerances of table III. Similar movements of each linear axis shall be programmed at varied feed rates to test for slide overshoot at accelerated feed rates without programmed deceleration. One pass shall be programmed at feed rates nearest to 10, 20, and 100 percent of maximum linear feed rate. Overshoot shall not exceed 0.003 inches at any feed rate up to 20 percent of the maximum. Maximum overshoot shall not exceed 0.005 inches. Stops shall be programmed appropriately for inspection of control positioning accuracy. Positioning tests may be part of the same tape program as 4.8.2.

4.8.4.2 Tapping test. Pilot drill and tap at programmed locations, four holes each with finished threads to accept 8-32 and 3/4-10 screws. Finished holes shall accept standard 8-32 or 3/4-10 screws freely and threads shall meet Class 3 standards for internal threads as set forth in FED-STD-H28.

4.8.4.3 Boring test. Pilot drill, semi-finish bore, and finish bore not less than 10 holes in a selected pattern covering an area not less than 13 inches in diameter. Four of the holes shall be spaced on 7 inch centers. Different tools shall be used for semi-finish and finish boring operations. All holes may be pilot drilled in sequence, but semi-finish and finish boring of each hole shall be completed with necessary tool changes before beginning boring operations of successive holes. Each address shall be to the fourth decimal place with digits other than zero. Finished hole size shall range from 1.125 to 2.750 inches. The resulting hole location and hole-size errors shall not exceed the permissible tolerance of table II for boring.

4.8.4.4 Straight-line milling test. The straight-line milling test shall be a programmed square or rectangle, not less than 12" per side with sides parallel to the travel along X and Y axes. Machines shall make a plunge cut along one longitudinal side cut. Plunge angle shall be not less than 2 degrees. The resulting cuts shall be checked for dimensional accuracy, squareness, parallelism, flatness, angularity, and corner locations to verify the accuracy required in table II for straight-line milling.

4.8.4.5 Circle contouring test. A circle shall be programmed and milled within the outline of the square cut required in 4.8.4.4. The cut shall be made at constant Z axis depth of one-half inch and by using the shortest programming sequence within the capability of the MCU. The resulting cut shall be checked for dimensional accuracy, roundness, surface finish, and angularity to verify the accuracy required in table II for circle contouring.

4.8.4.6 Straight-line contouring test. A square or an equilateral triangle shall be programmed and milled within the circle cut and described by 4.8.4.5. at a constant depth of one-half inch. The reference line for one side of the square or triangle shall be 10 to 20 angular degrees canted to the X axis. The resulting cut shall be checked for dimensional accuracy, squareness and angularity to verify the accuracy required in table II for straight-line contouring.

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4.8.4.7 Maximum horsepower test. Using a workpiece of SAE 1020 steel, a straight cut of not less than 12 inches in length shall be made with a face mill or shell end mill. Diameter of the cutter and spindle speed shall be determined by the supplier to be suitable for making the cut with specified chip load and peripheral cutter speeds. The depth of the cut shall be as required to load the spindle motor to its maximum rated horsepower. Chip load shall be 0.010 ± 0.002 inch and peripheral cutter speed shall not exceed 300 feet per minute. The cutting action shall be smooth and even and the finished workpiece shall show no evidence of tool chatter. In addition, there shall be no evidence of overheating of the spindle drive motor or the applicable feed motor(s).

4.8.4.8 Test program verification. Unless otherwise specified (see 6.2), while using one or more test set-ups devised by the supplier, the machining test program (tape) required under postprocessor validation (see 3.4.14.1), shall be run on the machining center to verify its ability to receive and perform all program functions without malfunction or error. The test piece(s) shall be either steel, cast iron, or aluminum, while the tooling shall be selected by the supplier as the most suitable for the particular operation being performed.

4.8.4.9 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.2.6. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.8.6 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The numerically controlled machines covered by this specification are intended for use in any production shop where drilling, boring, tapping, reaming, and milling operations by numerical control of the machine are required. The machine may be used to produce prototype parts from a taped processor to produce production items repetitively from a proven process, to accuracies within the capabilities of the machine.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Style and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify means of accomplishing Z axis movement, if different (see 3.2).

f. System of units for measuring and indicating devices (see 3.2.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

i Additional environmental requirements, if required (see 3.2.6).

j. Painting, if different (see 3.3.5).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

l. Specify the number and configuration of T-slots, if different (see 3.4.4).

m. Sub-plate and desired characteristics, if required (see 3.4.4).

n. Specify special characteristics for spindle noses and bores, if required (see 3.4.5.1).

o. Electrical system, if different (see 3.4.11).

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p. Initial voltage, if different (see 3.4.11).

q. Motors, if different (see 3.4.11.1).

r. Axis motor(s) and motor controls, if different (see 3.4.11.2).

s. Worklight, if not required (see 3.4.12).

t. Hour meter, if required (see 3.4.13).

u. Specify type of input media reader, if different (see 3.4.14.b).

v. Specify programmable software travel limits, if required (see 3.4.14.f).

w. Specify zone inhibit programming, if required (see 3.4.14.g).

x. Axis scaling, if required (see 3.4.14.h).

y. Specify polar coordinate programming, if required (see 3.4.14.i).

z. Specify radius programming, if required (see 3.4.14.k).

aa. Specify inverse time feed rate programming, if required (see 3.4.14.n).

bb. Specify parametric programming, if required (see 3.4.14.o).

cc. Specify helical interpolation, if required (see 3.4.14.r).

dd. Specify coordinate system plane rotation, if required (see 3.4.14.t).

ee. Specify either EIA Standards RS-232, RS-449 or both, (see 3.4.14.y).

ff. Specify software for rotary axis, if required (see 3.4.14.z).

gg. Specify number of axis to be CNC controlled, (see 3.4.14.cc).

hh. Specify tool life management, if required (see 3.4.14.dd).

ii. Specify multiquadrant circular interpolation, if required (see 3.4.14.jj).

jj. Specify the amount of standby battery backup power (in hours), if different (see 3.4.14.kk).

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kk. Specify the amount of part program storage (in feet), if different, (see 3.4.14.11).

ll. Specify type of part program storage media, (see 3.4.14.11).

mm. Postprocessor, specify computer name and model, form and format if required (see 3.4.14.1).

nn. Tape punch unit, if required (see 3.4.14.2).

oo. Diskette input, if required (see 3.4.14.3).

pp. If BCL is required, specify the function sets of EIA Standard RS-494 that are applicable to the machine being procurred (see 3.4.14.4) (NOTE: SHOULD ASSISTANCE BE RQUIRED IN COMPLETING FUNCTION SET REQUIREMENTS, CONTACT DIPEC-SSM)

qq. Punched tape data input, if different (see 3.4.14.5).

rr. Specify rotary axis designation as either A, B, or C, if required (see 3.4.14.6).

ss. Specify type of rotary table, contouring or positioning (see 3.4.14.6).

tt. Specify programmable increments for rotary axis, (in degrees), if required (see 3.4.14.6).

uu. Operator control panel functions, if different (see 3.4.14.7).

vv. Pendant control panel, if required (see 3.4.14.7.1).

ww. Specify pendant control functions, if different (see 3.4.14.7.1).

xx. Specify size and capacities, if different (see 3.5).

yy. Specify physical size limitations, if different (see 3.5.1).

zz. Additional performance requirements, if different (3.7.1).

aaa. Specify and fully describe optional equipment, if required (see 3.9).

bbb. Lubrication plate, if required (see 3.11.1).

ccc. Information to be provided on lubrication plate, if different (see 3.11.1).

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ddd. Nameplate, if different (see 3.11.2).

eee. Hydraulic plate, if required (see 3.11.3).

fff. Hydraulic plate, if different (see 3.11.3).

ggg. Inspection conditions, if different (see 4.3).

hhh. First article inspection, if different (see 4.4).

iii. Quality conformance inspection, if different (see 4.5).

jjj. Acceptance test, if different (see 4.6).

kkk. Test program verification, if different (see 4.8.4.8).

lll. Additional performance test, if required (see 4.8.4.9).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.13	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language
3.13	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the

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number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)", of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification changes. Classification changes from the previous issue of this specification are as follows:

MIL-M-80255B

Size 9 --------- New

Style F -------- New

Style G -------- New

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional

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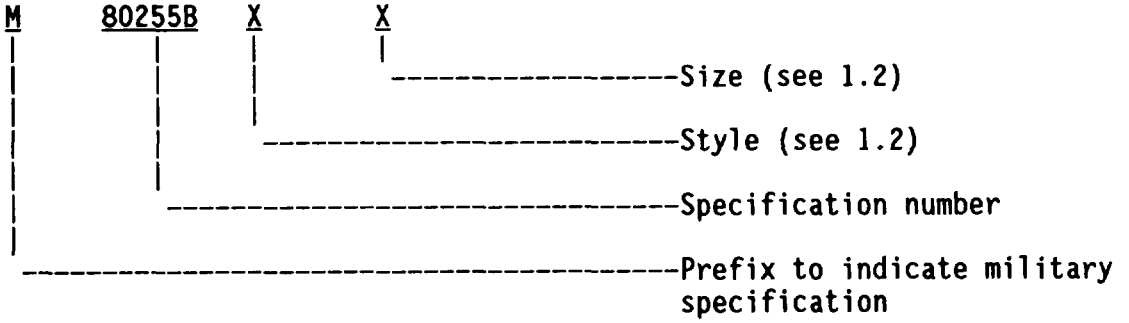
requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.14 Subject term (keyword) listing.

Base
Binary cutter location (BCL)
Column
Diskette
Interface
Machining head
Manual data input
Mill Rotary table
Saddle
Software
Spindle
Tool changer

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TABLE I Sizes and capacities.

Characteristics	Requirements			
	Size 1	Size 2	Size 3	Size 4
Spindle Motor horsepower	5	7	7	10
Tape controlled travel, inches				
X axis	30	30	50	40
Y axis	15	15	26	20
Z axis	8	10	10	12
Capacity (in SAE 1020 steel)				
Drill, dia., inches	1	1.25	2	1.75
Tap, dia., inches	1.25	1.25	2	1.50
Mill, cubic in. per min.	5	7	7	10
Machining head & spindle assembly				
Spindle speeds, number	inf.	inf.	inf.	inf.
Spindle speed range, RPM	100 to 2000	100 to 2000	45 to 3000	45 to 1500
Spindle feeds, number	inf.	inf.	inf.	inf.
Spindle feed rate range, in. per min.,	0.5 to 90	0.5 to 100	0.2 to 100	0.5 to 200
Spindle travel (z axis), in.	8	10	10	12
Spindle rapid traverse rate, in. per min.	80	100	200	200
Machining head or worktable vertical travel (z axis), in.				
Machining head or worktable vertical rapid traverse rate, in. per min.	80	100	200	200
Worktable				
Work mounting surface, length x width, in.	38 x 16	38 x 16	54 x 26	44 x 20
Load capacity, lbs.	1000	2000	2500	2500
Worktable (saddle or column) table, in.				
X axis	30	30	50	40
Y axis	15	15	26	20
Rapid traverse rate, in. per min.	120	150	200	200
Milling feeds, number	inf.	inf.	inf.	inf.
Milling feed rate range, in. per min.	0.5 to 90	0.5 to 100	0.2 to 100	0.5 to 200
Tool changer				
Chip to chip time (successive operations two different tools in the ready position), seconds	12	12	12	12

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TABLE I Sizes and capacities, continued

Characteristics	Requirements				
	Size 5	Size 6	Size 7	Size 8	Size 9
Spindle Motor Horsepower	10	10	15	20	40
Tape controlled travel, inches					
x axis	50	80	54	54	80
y axis	26	26	28	28	40
z axis	12	12	14	24	40
Capacity (in SAE 1020 steel)					
Drill, dia., inches	1.75	1.75	2	2	4
Tap, dia., inches	1.50	1.75	2	2	4
Mill, cubic in. per min.	10	10	15	20	40
Machining head & spindle assembly					
Spindle speeds, number	inf.	inf.	inf.	inf.	inf.
Spindle speed range, RPM	70 to 3000	70 to 3000	30 to 3000	30 to 3000	40 to 4000
Spindle feeds, number	inf.	inf.	inf.	inf.	inf.
Spindle feed rate range, in. per min.,	0.5 to 200	0.5 to 200	0.5 to 200	0.5 to 150	0.1 to 300
Spindle travel (z axis), in.	12	12	14	24	---
Spindle rapid traverse rate, in. per min.	200	200	200	200	---
Machining head or worktable vertical travel (z axis), in.	---	---	---	---	40
Machining head or worktable vertical rapid traverse rate, in. per min.	200	200	200	200	300
Worktable					
Work mounting surface, length x width, in.	54 x 26	80 x 26	60 x 28	60 x 28	80 x 40
Load capacity, lbs.	3000	3500	3500	3500	---
Worktable (saddle or column) table, in.					
x axis	50	80	54	54	80
y axis	26	26	28	28	40
Rapid traverse rate, in. per min.	200	200	200	200	400
Milling feeds, number	inf.	inf.	inf.	inf.	inf.
Milling feed rate range, in. per min.	0.5 to 200	0.5 to 200	0.5 to 200	0.5 to 150	0.1 to 300
Tool changer					
Chip to chip time (successive operations two different tools in the ready position), seconds	12	12	12	12	17

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TABLE II Performance accuracy.

Machining Operation	Accuracy (permissible tolerance)
Drilling	
Depth of hole	±0.001"
Tapping	
Threaded holes shall be Class III in accordance with FED-STD-H28	
Boring and Counterboring	
Hole location, based on centerline distance of bored holes, 2 holes, 18" apart	±0.002"
Hole size, maximum spread, hole to hole	±0.002"
Depth of counterbore	±0.001"
Milling (straight-line)	
Squareness of sides	0.002"/ft.
Parallelism of opposing sides	0.002"/ft.
Flatness	±0.002"/3 ft.
Angularity of ramp cuts	±0.002"/ft.
Corner location, deviation from programmed points	±0.001"
Contouring (circle)	
Roundness	
Diameters measured at 30 degree intervals	±0.002"
Radii measured at 30 degree intervals	±0.002"
Surface finish, maximum roughness 64 RMS	
Angularity	
Difference between programmed and actual points of tangency of circles to straight surfaces	0.002"
Contouring (straight-line)	
Squareness of sides	0.002"/ft.
Angularity of sides	0.002"/ft.

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TABLE III. Alignment and accuracy.

Component Characteristic	Permissible Tolerance
Worktable:	
Flatness of work mounting surface	0.001"/ft.
Squareness of work mounting surface to spindle axis, in any 6" sweep radius	0.002"/ft.
Motion in the x axis, rise and fall with respect to spindle	0.001"/ft.
Motion in the y axis, rise and fall with respect to the spindle	0.001"/ft.
Parallelism of T-slots with x axis movement	0.0005"/ft.
Squareness of T-slots with y axis movement (slide travel squareness)	0.0005"/ft.
Spindle:	
Radial runout:	
1" from spindle nose	0.0003" TIR
6" from spindle nose	0.001" TIR
Squareness of spindle movement to the work mounting surface of the worktable per 6" of travel	0.0005"
Numerical control system:	
x & y axes positioning and repeatability accuracy:	
absolute positioning	±0.001"/ft.
position repeatability	±0.0005"
z axis positioning and repeatability accuracy:	
absolute positioning	±0.001"/ft.
positioning repeatability	±0.0005"
a axis (when furnished) positioning and repeatability accuracy:	±10 sec./ft.
positioning repeatability	±5 sec.
Axis overshoot:	
20 % of maximum feed rate	0.003"
Maximum feed rate	0.005"

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AL
Navy - SH, AS
Air Force - 84
DLA - GS

User activities:
Navy - OS, AS

Preparing activity:
DLA - IP

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-80255B	900913

3. DOCUMENT TITLE
Machining Centers, Vertical, Single Spindle, Computer Numerical Control (CNC)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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